

3 Population and samples

Subject content	Additional information
3.1 Know both simple (without replacement) and unrestricted (with replacement) random samples.	Students should know that for a random sample of size n: • every member of the population is equally likely to be included • all subsets of the population of size n must be possible or that • every possible sample of size n must be equally likely to occur.
3.2 Know how to obtain a random sample using random numbers tables or random numbers generated on a calculator.	
3.3 Evaluate the practical application of random and non-random sampling techniques: simple random, systematic, cluster, judgmental and snowball, including the use of stratification (in proportional and disproportional ratios) prior to sampling taking place.	Students should appreciate that snowball sampling can be used to reach populations that are difficult to sample when using other sampling methods e.g. drug users.
3.4 Know the advantages and limitations of sampling methods.	

3 Population and samples (continued)

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3.5 Make reasoned choices with reference to the context in which the sampling is to take place. Examples include, but are not limited to: market research, exit polls, experiments and quality assurance.	
3.6 Understand the practical constraints of collecting unbiased data.	